

CSA 0378-DATA STRUCTURES

23. PRIM'S ALGORITHM

main.c	Run	Output
<pre>1 #include <stdio.h> 2 3 int main() { 4 int graph[4][4] = { 5 {0,1,4,0}, 6 {1,0,2,5}, 7 {4,2,0,1}, 8 {0,5,1,0} 9 }; 10 11 int dist[4] = {0,999,999,999}; 12 int visited[4] = {0}; 13 int i,j,min,u; 14 15 for(i=0;i<4;i++) { 16 min = 999; 17 18 for(j=0;j<4;j++) 19 if(!visited[j] && dist[j] < min) { 20 min = dist[j]; 21 u = j; 22 } 23 24 visited[u] = 1; 25 26 for(j=0;j<4;j++) 27 if(graph[u][j] && dist[u] + graph[u][j] < dist[j]) 28 dist[j] = dist[u] + graph[u][j]; 29 } 30 31 printf("Distances:\n"); 32 for(i=0;i<4;i++) 33 printf("%d ", dist[i]); 34 35 return 0; 36 }</pre>		Distances: 0 1 3 4 === Code Execution Successful ===